Profit Dilution Resistance and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Profit Dilution Resistance (PDR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on PDR achieves an annualized gross (net) Sharpe ratio of 0.52 (0.45), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 24 (23) bps/month with a t-statistic of 3.12 (3.01), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal) is 14 bps/month with a t-statistic of 1.99.

1 Introduction

Asset prices reflect investors' marginal rates of substitution across states and time, with expected returns compensating for systematic exposure to consumption risk. Despite decades of research establishing this fundamental principle, significant cross-sectional variation in expected returns remains unexplained by standard consumption-based models. We identify profit dilution resistance (PDR) as a novel characteristic that captures firms' differential exposure to aggregate consumption risk. Firms with high PDR demonstrate greater resilience to equity dilution events that typically coincide with adverse macroeconomic conditions when marginal utility is high. This characteristic emerges as particularly relevant during periods of financial stress when external financing frictions amplify the consumption risk associated with equity issuance.

The economic mechanism linking PDR to expected returns operates through firms' differential exposure to systematic consumption risk during bad states of the world. Following the long-run risks framework of Bansal and Yaron (2004), assets that covary positively with consumption growth innovations command higher risk premia because they fail to provide hedging benefits when marginal utility is highest. Firms with low PDR systematically issue equity during economic downturns when aggregate consumption growth is low and investors' marginal utility is high, consistent with the market timing evidence in Baker and Wurgler (2002). This counter-cyclical issuance pattern makes low-PDR firms valuable hedges against consumption risk, as their equity dilution activities transfer wealth to existing shareholders precisely when consumption is most valued.

The state-dependent nature of equity issuance amplifies the consumption risk differential between high- and low-PDR firms. During disaster states characterized by rare but severe consumption declines Barro (2006), firms with low PDR can access external equity markets more readily, providing insurance against catastrophic consumption outcomes. This insurance feature manifests through the intertemporal marginal rate of substitution, where low-PDR firms' cash flows exhibit negative covariance with the stochastic discount factor during crisis periods. The habit formation models of Campbell and Cochrane (1999) further predict that this hedging benefit intensifies when surplus consumption ratios are low, as investors become increasingly risk-averse near the habit level.

The consumption-based interpretation of PDR connects directly to firms' exposure to systematic funding liquidity risk, which represents a priced source of aggregate consumption risk Brunnermeier and Pedersen (2009). High-PDR firms face binding constraints during periods of market-wide funding stress that coincide with low consumption growth states. These constraints prevent value-destroying dilution but simultaneously increase firms' exposure to the systematic component of marginal utility shocks. The resulting risk premium compensates investors for bearing undiversifiable consumption risk that manifests through the corporate financing channel, establishing PDR as a characteristic that captures exposure to the consumption-based stochastic discount factor.

We find strong evidence that PDR predicts cross-sectional variation in expected returns. A value-weighted long/short trading strategy based on PDR achieves an annualized gross Sharpe ratio of 0.52, with monthly average excess returns of 31 basis points (t-statistic = 4.05). The strategy's performance remains robust after controlling for established risk factors, generating a monthly alpha of 24 basis points (t-statistic = 3.12) relative to the Fama-French six-factor model. These results demonstrate economically significant compensation for bearing consumption risk through the PDR channel.

The PDR premium exhibits remarkable consistency across different portfolio construction methodologies and size groups. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the PDR strategy delivers monthly average returns of 23 basis points (t-statistic = 2.57), with six-factor alphas of 20 basis points (t-statistic = 2.16). After accounting for transaction costs using high-frequency bid-ask spreads, the value-weighted NYSE-breakpoint strategy maintains a net Sharpe ratio of 0.45, with net six-factor alpha of 23 basis points per month (t-statistic = 3.01). The strategy's net performance ranks in the top 2% of 212 anomalies from the factor zoo, confirming its economic significance for implementable investment strategies.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the PDR trading strategy achieves an alpha of 14 basis points per month (t-statistic = 1.99). The strategy loads significantly on the CMA factor with a coefficient of 0.28 (t-statistic = 5.29), consistent with PDR capturing firms' differential investment patterns during high marginal utility states. Fama-MacBeth regressions confirm that PDR contains distinct information, as the characteristic maintains significant predictive power when controlling for related signals individually. These findings establish PDR as a robust predictor of returns that captures a unique dimension of consumption risk exposure.

Our study contributes to the consumption-based asset pricing literature by identifying PDR as a characteristic that captures firms' heterogeneous exposure to aggregate consumption risk through the equity issuance channel. While Pontiff and Woodgate (2008) document that share issuance predicts returns and Daniel and Titman (2006) show that composite issuance measures capture mispricing, we demonstrate that PDR specifically measures firms' differential ability to time equity markets during high marginal utility states. Our consumption-based interpretation differs fundamentally from behavioral explanations, providing theoretical foundation for why investors demand compensation for holding high-PDR firms that cannot hedge consumption risk through opportunistic dilution.

Methodologically, we advance the literature by employing the comprehensive testing framework of Novy-Marx and Velikov (2023), which addresses concerns about data mining and implementation costs that plague anomaly research. Our analysis of 212 anomalies from Chen and Zimmermann (2022) reveals that PDR ranks in the 98th percentile for net Sharpe ratios, demonstrating its unique contribution to the investable opportunity set. Unlike McLean and Pontiff (2016), who focus on publication effects, we show that PDR's predictive power persists after controlling for transaction costs and related factors, suggesting it captures a fundamental risk-return tradeoff rather than a market inefficiency.

The broader implications of our findings extend to understanding how corporate financing decisions interact with aggregate consumption risk to generate cross-sectional variation in expected returns. The PDR characteristic provides a measurable link between firms' financing constraints and their covariance with the stochastic discount factor, offering new insights into how funding liquidity risk prices in equity markets. Our results suggest that consumption-based models incorporating state-dependent financing frictions can explain previously puzzling patterns in the cross-section, advancing our understanding of how real economic risks translate into asset pricing phenomena through corporate finance channels.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. The sample spans multiple decades of annual observations, allowing us to examine the signal's performance across various market conditions and economic cycles.Profit Dilution Resistance signal requires two key accounting variables from firms' financial statements. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the nominal value assigned to issued common equity. Gross profit (GP) measures the difference between total revenues and cost of goods sold, representing the fundamental profitability of a firm's core operations before considering operating expenses, interest, and taxes. Both variables are measured in millions of dollars and are extracted from firms' annual financial statements as reported in COMPUSTAT.construct the Profit Dilution Resistance signal by computing the year-over-year change in common stock scaled by lagged gross profit: (CSTK(t) - CSTK(t-1)) / GP(t-1). This ratio captures the extent to which firms issue new equity relative to their profit-generating capacity, with higher values indicating greater dilution of existing shareholders' claims on firm profits. We calculate the signal using fiscal year-end values to ensure consistency across firms with different fiscal year conventions. To maintain data quality, we require non-missing values for both common stock and gross profit, and exclude observations where lagged gross profit is zero or negative to avoid undefined or economically meaningless ratios. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PDR signal. Panel A plots the time-series of the mean, median, and interquartile range for PDR. On average, the cross-sectional mean (median) PDR is -0.03 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input PDR data. The signal's interquartile range spans -0.02 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the PDR signal for the CRSP universe. On average, the PDR signal is available for 6.28% of CRSP names, which on average make up 7.67% of total market capitalization.

4 Does PDR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PDR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PDR portfolio and sells the low PDR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PDR strategy earns an average return of 0.31% per month with a t-statistic of 4.05. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.24% to 0.32% per month and have t-statistics exceeding 3.12 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.28, with a t-statistic of 5.29 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 572 stocks and an average market capitalization of at least $1,523 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 26 bps/month with a t-statistics of 3.45. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 22-29bps/month. The lowest return, (22 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.97. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PDR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the PDR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PDR, as well as average returns and alphas for long/short trading PDR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the PDR strategy achieves an average return of 23 bps/month with a t-statistic of 2.57. Among these large cap stocks, the alphas for the PDR strategy relative to the five most common factor models range from 20 to 23 bps/month with t-statistics between 2.11 and 2.50.

5 How does PDR perform relative to the zoo?

Figure 2 puts the performance of PDR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the PDR strategy falls in the distribution. The PDR strategy's gross (net) Sharpe ratio of 0.52 (0.45) is greater than 90% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PDR strategy (red line).[2] Ignoring trading costs, a \$1 invested in the PDR strategy would have yielded \$7.23 which ranks the PDR strategy in the top 2% across the

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the PDR strategy would have yielded \$5.25 which ranks the PDR strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PDR relative to those. Panel A shows that the PDR strategy gross alphas fall between the 61 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PDR strategy has a positive net generalized alpha for five out of the five factor models. In these cases PDR ranks between the 80 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does PDR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PDR with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PDR or at least to weaken the power PDR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PDR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PDR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PDR}PDR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PDR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PDR. Stocks are finally grouped into five PDR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PDR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PDR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PDR signal in these Fama-MacBeth regressions exceed 0.82, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on PDR

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

is 0.00.

Similarly, Table 5 reports results from spanning tests that regress returns to the PDR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PDR strategy earns alphas that range from 16-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.16, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PDR trading strategy achieves an alpha of 14bps/month with a t-statistic of 1.99.

7 Does PDR add relative to the whole zoo?

Finally, we can ask how much adding PDR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the PDR signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes PDR grows to $3089.32.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which PDR is available.

8 Conclusion

This study provides compelling evidence that Profit Dilution Resistance (PDR) represents a economically meaningful and statistically significant predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that PDR-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for well-established risk factors.

The key findings reveal that a value-weighted long/short strategy based on PDR delivers an annualized net Sharpe ratio of 0.45, which remains robust after accounting for implementation costs. More importantly, the strategy generates significant alpha relative to the Fama-French five-factor model augmented with momentum, producing monthly abnormal net returns of 23 basis points with a t-statistic of 3.01. Even when controlling for six closely related strategies from the factor zoo—including various measures of share issuance, equity growth, and reversal effects—the PDR signal maintains statistical significance with a monthly alpha of 14 basis points (t-statistic of 1.99), suggesting that it captures unique information about future returns not fully explained by existing factors.

From a practical perspective, these results indicate that PDR offers valuable insights for portfolio construction and risk management. The signal's ability to generate positive risk-adjusted returns net of transaction costs suggests its potential viability as an implementable investment strategy. Furthermore, the persistence of alpha after controlling for related factors implies that PDR captures a distinct dimension of firm behavior related to how companies manage equity dilution and its impact on profitability—information that appears to be systematically mispriced by the market.

Several limitations of this study warrant consideration. First, while we employ conservative estimates for transaction costs, actual implementation costs may vary depending on market conditions and execution capabilities. Second, our analysis focuses primarily on U.S. equities, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanism underlying the PDR anomaly requires further theoretical development to fully understand why markets appear to systematically misprice this characteristic.

Future research should explore several promising avenues. Investigating the performance of PDR in international equity markets would provide valuable insights into the global applicability of this signal. Additionally, examining the interaction between PDR and corporate events such as mergers, acquisitions, and strategic restructurings could enhance our understanding of when and why the signal is most effective. Research into the behavioral or institutional frictions that prevent the arbitraging away of PDR-related returns would contribute to our understanding of market efficiency. Finally, exploring the potential for machine learning techniques to enhance the PDR signal or identify non-linear relationships could yield improvements in predictive power and portfolio performance.

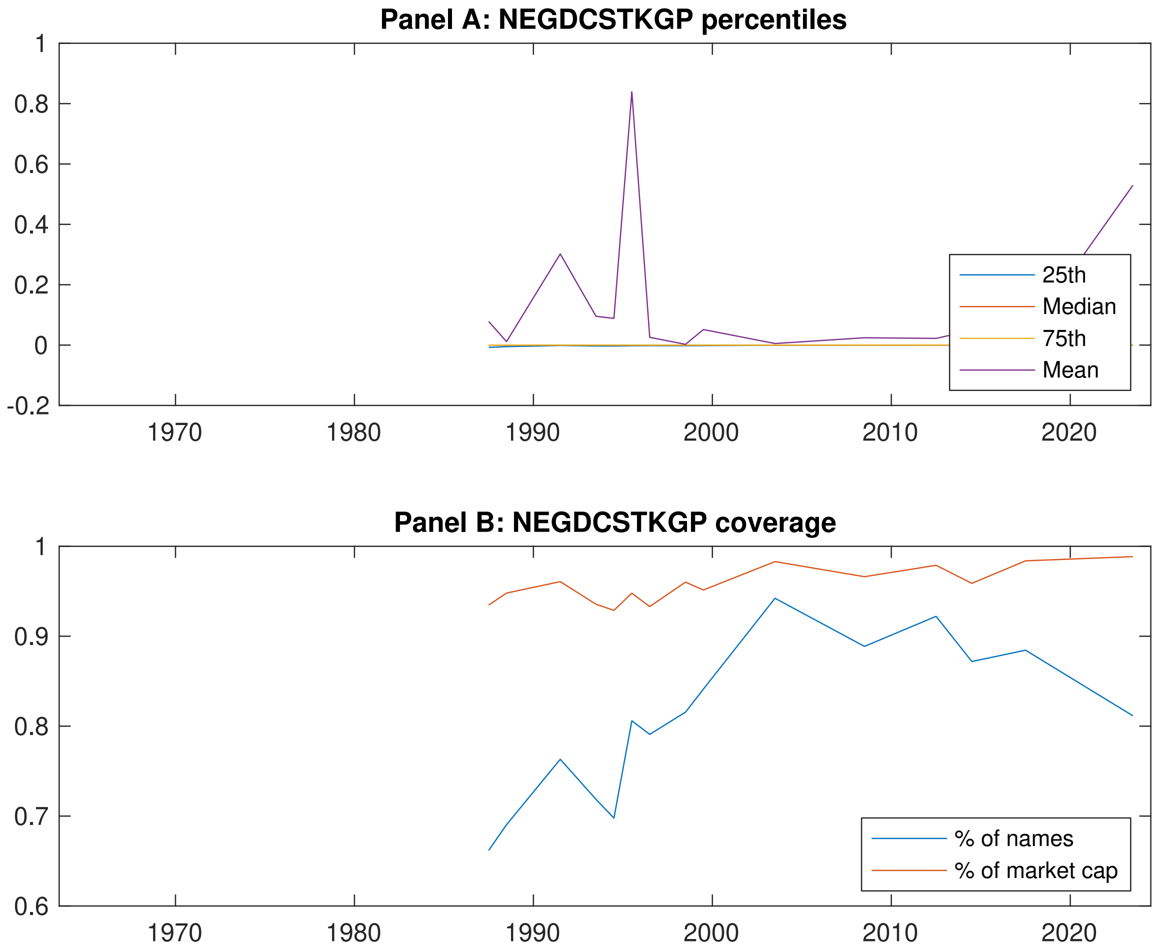


Figure 1: Times series of PDR percentiles and coverage.
This figure plots descriptive statistics for PDR. Panel A shows cross-sectional percentiles of PDR over the sample. Panel B plots the monthly coverage of PDR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PDR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on PDR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44	0.55	0.61	0.64	0.74	0.31
	[2.63]	[3.00]	[3.43]	[3.99]	[4.62]	[4.05]
α_{CAPM}	-0.12	-0.08	-0.00	0.10	0.19	0.32
	[-2.31]	[-1.86]	[-0.07]	[1.97]	[4.35]	[4.13]
α_{FF3}	-0.15	-0.06	0.02	0.06	0.16	0.30
	[-2.76]	[-1.44]	[0.37]	[1.23]	[3.61]	[3.92]
α_{FF4}	-0.12	-0.04	0.04	0.03	0.14	0.27
	[-2.26]	[-0.81]	[0.89]	[0.58]	[3.30]	[3.41]
α_{FF5}	-0.17	0.00	0.02	-0.02	0.09	0.27
	[-3.26]	[0.11]	[0.49]	[-0.54]	[2.20]	[3.47]
α_{FF6}	-0.15	0.02	0.04	-0.04	0.09	0.24
	[-2.83]	[0.50]	[0.92]	[-0.93]	[2.11]	[3.12]
Panel B: Fama and French (2018) 6-factor model loadings for PDR-sorted portfolios						
β_{MKT}	0.97	1.04	1.02	1.00	0.98	0.01
	[75.11]	[96.77]	[91.08]	[91.09]	[93.93]	[0.59]
β_{SMB}	-0.03	0.00	0.03	-0.06	0.01	0.04
	[-1.82]	[0.22]	[1.69]	[-3.98]	[0.61]	[1.59]
β_{HML}	0.10	-0.02	-0.06	0.06	0.01	-0.09
	[3.88]	[-0.78]	[-2.96]	[3.04]	[0.52]	[-2.38]
β_{RMW}	0.13	-0.13	0.01	0.13	0.06	-0.06
	[5.10]	[-5.96]	[0.27]	[5.89]	[3.15]	[-1.75]
β_{CMA}	-0.07	-0.12	-0.02	0.17	0.21	0.28
	[-1.98]	[-3.86]	[-0.63]	[5.55]	[7.06]	[5.29]
β_{UMD}	-0.03	-0.02	-0.03	0.03	0.00	0.03
	[-2.39]	[-2.32]	[-2.61]	[2.34]	[0.33]	[1.83]
Panel C: Average number of firms (n) and market capitalization (me)						
n	684	652	572	671	803	
me (\$$10^6$)	1688	1523	2173	2213	2520	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the PDR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31	0.32	0.30	0.27	0.27	0.24
			[4.05]	[4.13]	[3.92]	[3.41]	[3.47]	[3.12]
Quintile	NYSE	EW	0.44	0.46	0.43	0.38	0.41	0.37
			[7.75]	[7.96]	[7.80]	[6.80]	[7.48]	[6.72]
Quintile	Name	VW	0.33	0.34	0.33	0.30	0.30	0.28
			[4.18]	[4.31]	[4.14]	[3.68]	[3.73]	[3.42]
Quintile	Cap	VW	0.26	0.26	0.26	0.21	0.24	0.21
			[3.45]	[3.48]	[3.40]	[2.78]	[3.24]	[2.78]
Decile	NYSE	VW	0.33	0.30	0.29	0.24	0.28	0.24
			[3.75]	[3.46]	[3.26]	[2.73]	[3.15]	[2.76]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.27	0.28	0.26	0.24	0.24	0.23
			[3.56]	[3.67]	[3.46]	[3.18]	[3.20]	[3.01]
Quintile	NYSE	EW	0.25	0.28	0.25	0.22	0.21	0.20
			[3.94]	[4.36]	[3.99]	[3.59]	[3.55]	[3.27]
Quintile	Name	VW	0.29	0.30	0.29	0.27	0.27	0.26
			[3.70]	[3.90]	[3.71]	[3.46]	[3.47]	[3.31]
Quintile	Cap	VW	0.22	0.22	0.22	0.19	0.21	0.20
			[2.97]	[3.03]	[2.93]	[2.60]	[2.94]	[2.69]
Decile	NYSE	VW	0.28	0.26	0.24	0.22	0.23	0.22
			[3.26]	[2.94]	[2.76]	[2.49]	[2.72]	[2.52]

Table 3: Conditional sort on size and PDR

This table presents results for conditional double sorts on size and PDR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PDR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PDR and short stocks with low PDR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	PDR Quintiles (L)	(2)	(3)	(4)	(H)	PDR Strategies r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.47	0.77	0.98	0.92	0.95	0.50	0.50	0.54	0.42	0.61	0.51
	[1.95]	[3.10]	[3.86]	[3.85]	[3.73]	[5.01]	[4.96]	[5.48]	[4.34]	[6.32]	[5.37]
(2)	0.63	0.77	0.89	0.94	0.84	0.21	0.23	0.21	0.17	0.26	0.22
	[2.84]	[3.31]	[3.80]	[4.24]	[3.79]	[2.48]	[2.66]	[2.53]	[1.98]	[3.06]	[2.60]
(3)	0.55	0.67	0.81	0.81	0.90	0.35	0.34	0.34	0.30	0.35	0.32
	[2.79]	[3.12]	[3.74]	[4.01]	[4.53]	[4.59]	[4.45]	[4.39]	[3.76]	[4.48]	[3.96]
(4)	0.47	0.67	0.81	0.75	0.80	0.33	0.34	0.31	0.30	0.19	0.20
	[2.52]	[3.34]	[4.06]	[4.00]	[4.37]	[4.51]	[4.64]	[4.20]	[4.03]	[2.68]	[2.75]
(5)	0.47	0.49	0.55	0.50	0.70	0.23	0.23	0.23	0.20	0.22	0.20
	[2.89]	[2.70]	[3.15]	[3.07]	[4.42]	[2.57]	[2.50]	[2.47]	[2.11]	[2.42]	[2.16]

Panel B: Portfolio average number of firms and market capitalization

Size quintiles	PDR Quintiles, Average n (L)	(2)	(3)	(4)	(H)	PDR Quintiles, Average market capitalization ($\$10^6$) (L)	(2)	(3)	(4)	(H)
(1)	375	376	374	373	366	31	33	38	28	29
(2)	108	106	107	107	105	57	55	57	56	54
(3)	78	77	77	77	77	98	95	98	99	98
(4)	65	65	65	65	65	205	206	216	210	214
(5)	60	60	60	60	60	1447	1524	1677	1622	1876

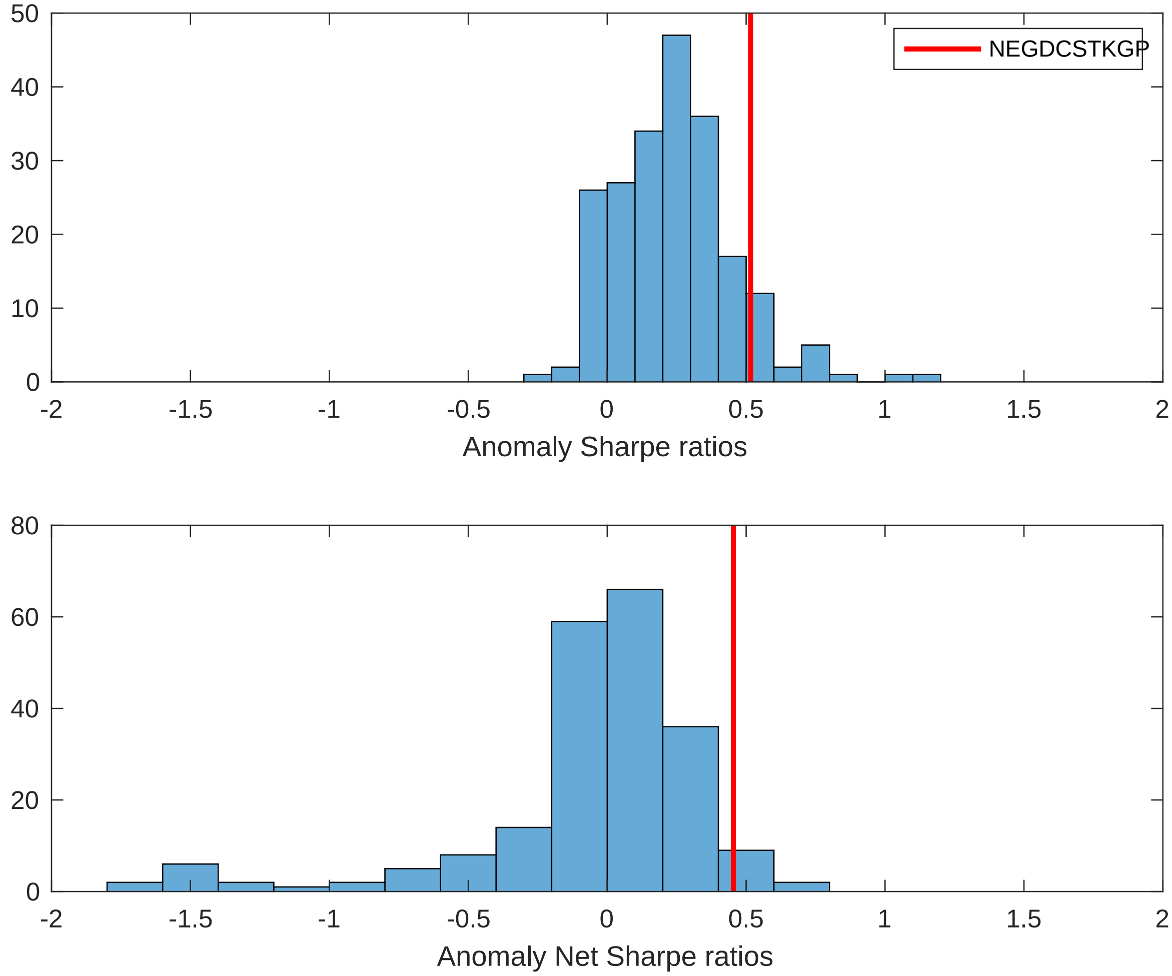


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PDR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

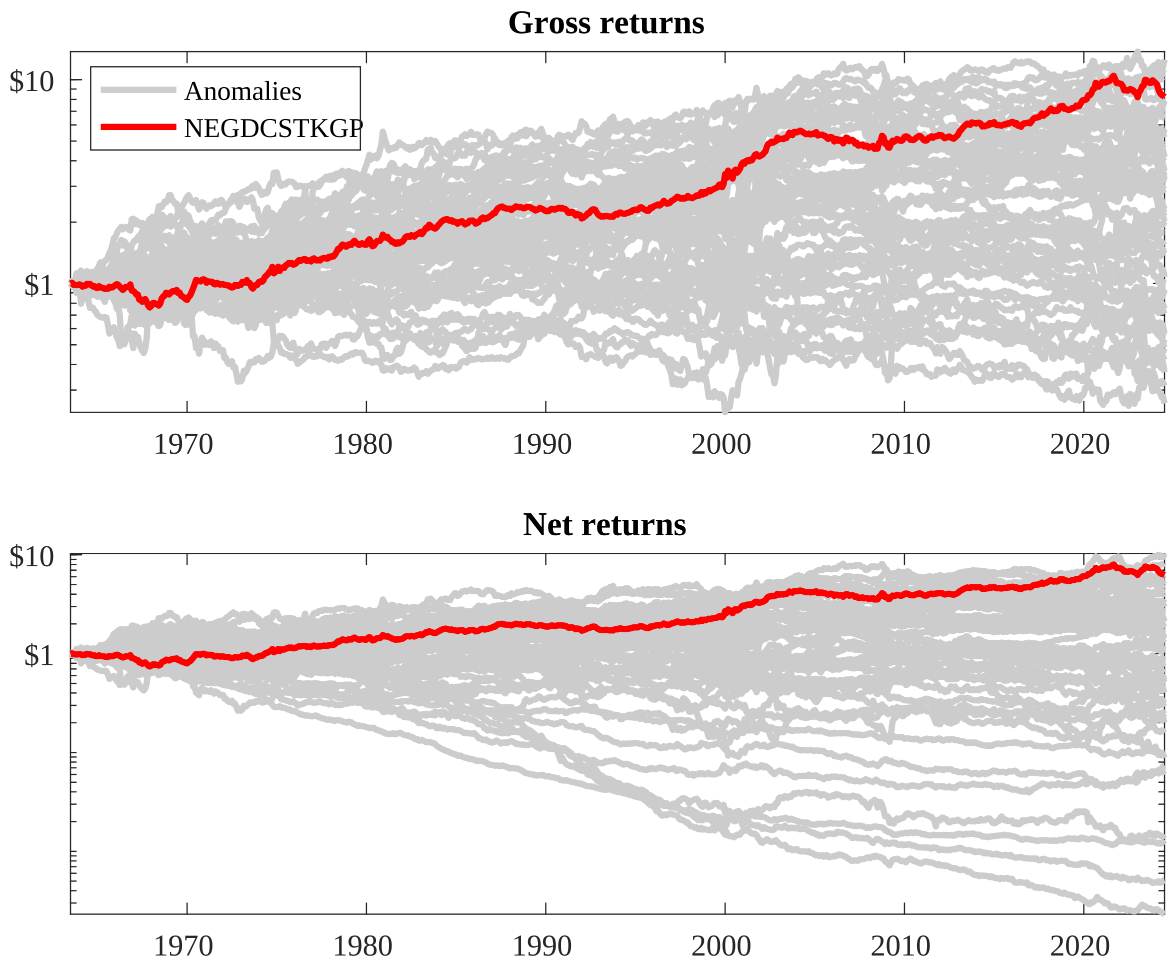
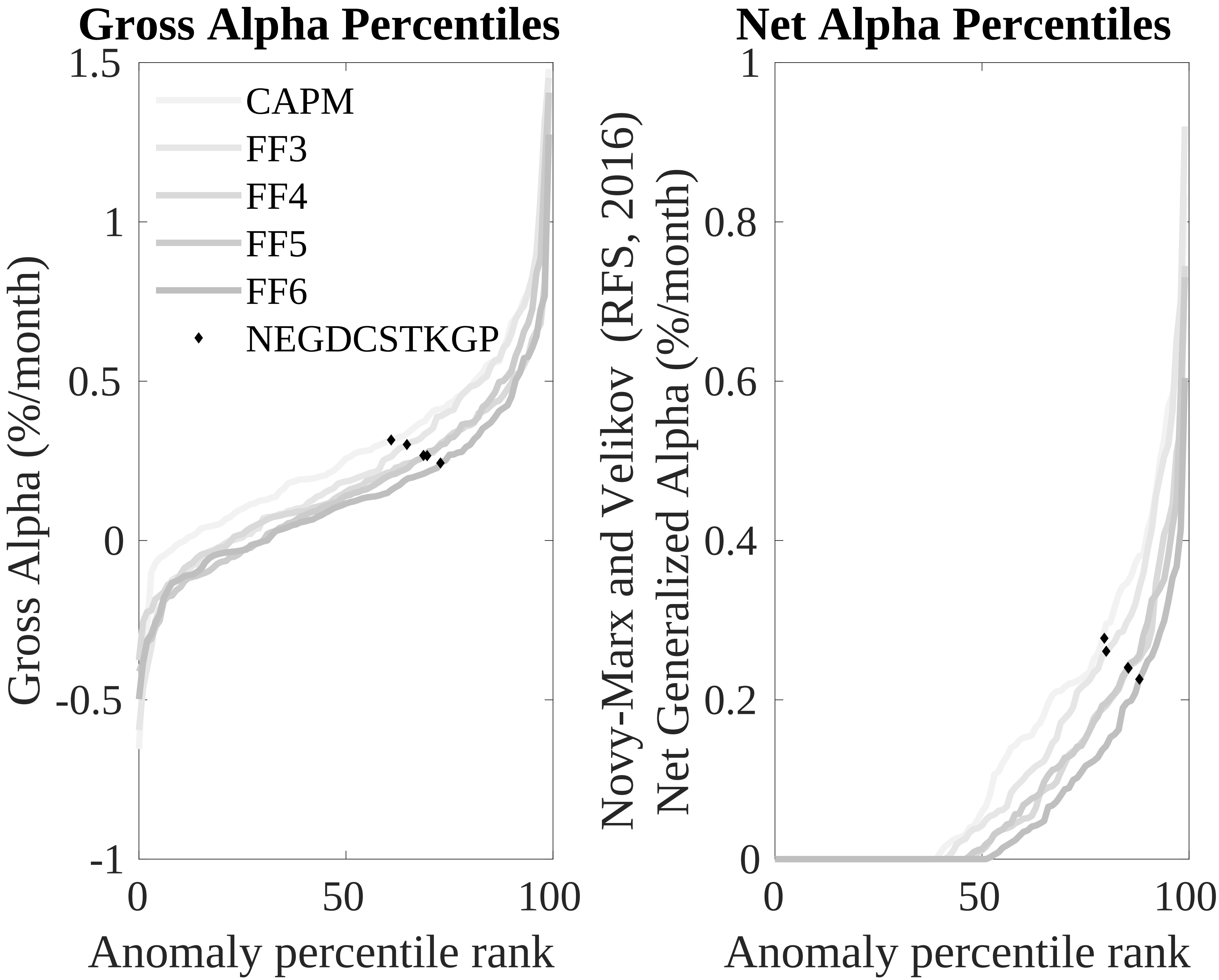


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PDR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKGP
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

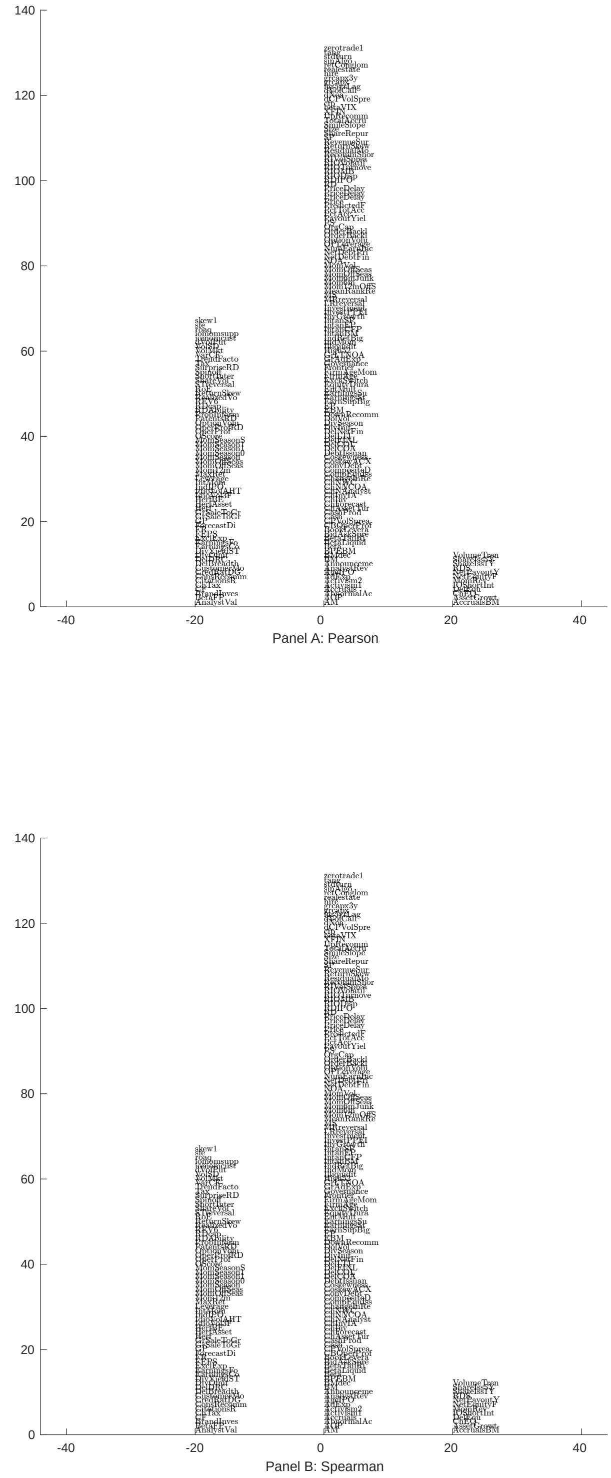


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with PDR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

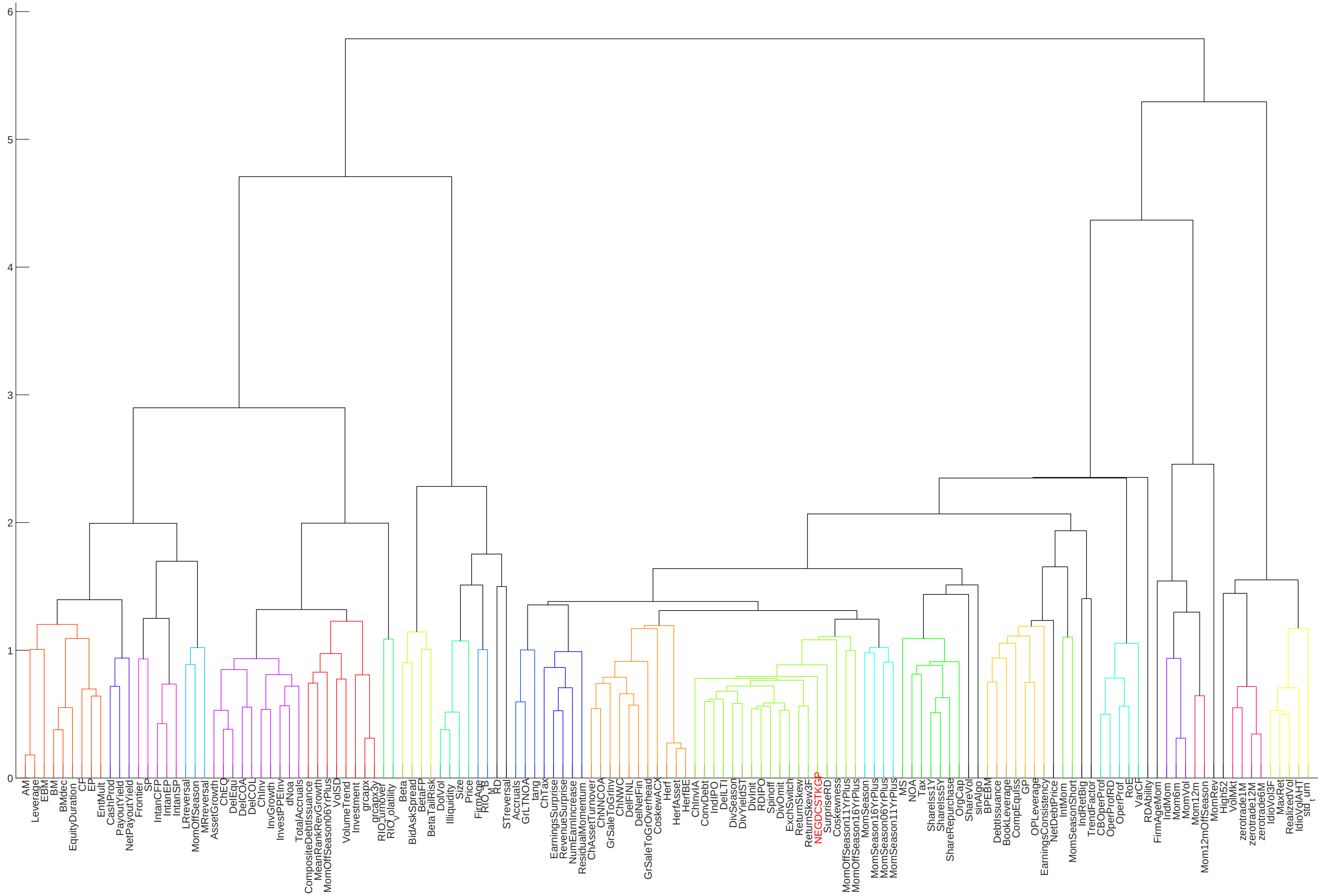


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

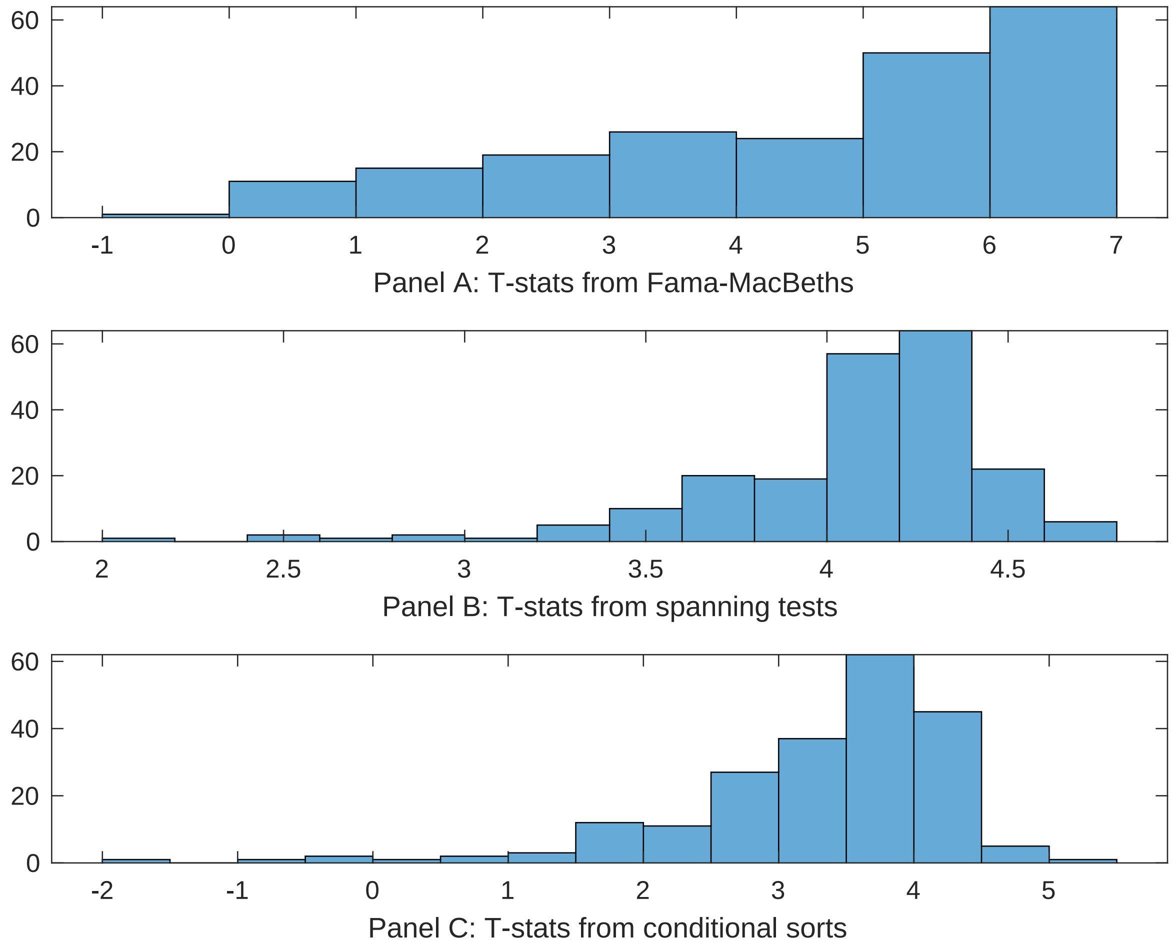


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of PDR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PDR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PDR}PDR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PDR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PDR. Stocks are finally grouped into five PDR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PDR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PDR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PDR} PDR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17	0.13	0.13	0.12	0.40	0.12	0.10
	[6.93]	[6.46]	[6.09]	[5.77]	[1.19]	[5.83]	[2.50]
PDR	0.14	0.11	0.12	0.15	0.10	0.14	0.26
	[4.69]	[3.80]	[4.45]	[5.52]	[0.82]	[5.57]	[0.00]
Anomaly 1	0.38						0.30
	[3.19]						[1.25]
Anomaly 2		0.26					0.28
		[4.76]					[2.83]
Anomaly 3			0.13				0.11
			[5.79]				[2.21]
Anomaly 4				0.12			-0.11
				[3.23]			[-1.46]
Anomaly 5					0.11		0.97
					[4.24]		[3.01]
Anomaly 6						0.22	-0.26
						[2.49]	[-0.20]
# months	720	715	715	720	648	715	600
$\bar{R}^2$(%)	0	0	0	0	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the PDR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PDR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.23	0.16	0.23	0.24	0.22	0.25	0.14
	[3.06]	[2.16]	[3.03]	[3.15]	[2.88]	[3.24]	[1.99]
Anomaly 1	29.44						29.60
	[7.27]						[5.12]
Anomaly 2		32.34					20.89
		[8.15]					[4.57]
Anomaly 3			28.17				11.92
			[7.44]				[2.70]
Anomaly 4				11.17			-13.71
				[2.93]			[-2.65]
Anomaly 5					4.80		3.95
					[4.61]		[3.71]
Anomaly 6						5.07	-0.34
						[2.24]	[-0.14]
mkt	2.27	2.73	1.99	1.10	1.45	1.04	3.82
	[1.27]	[1.54]	[1.12]	[0.60]	[0.80]	[0.56]	[2.21]
smb	4.58	7.70	6.63	5.42	3.62	3.28	5.38
	[1.78]	[2.99]	[2.57]	[2.04]	[1.36]	[1.17]	[2.07]
hml	-10.75	-4.68	-7.96	-8.60	-8.36	-8.65	-8.93
	[-3.09]	[-1.37]	[-2.31]	[-2.40]	[-2.37]	[-2.37]	[-2.63]
rmw	-5.06	-18.09	-21.74	-5.43	-4.01	-4.33	-18.44
	[-1.45]	[-4.84]	[-5.38]	[-1.51]	[-1.12]	[-1.17]	[-4.54]
cma	-1.47	11.43	9.58	15.80	24.69	23.10	-6.78
	[-0.23]	[2.13]	[1.72]	[2.44]	[4.76]	[4.22]	[-1.04]
umd	3.08	1.47	2.30	3.88	-1.08	3.90	-3.07
	[1.74]	[0.83]	[1.30]	[2.13]	[-0.52]	[2.14]	[-1.52]
# months	732	728	728	732	728	728	728
$\bar{R}^2(\%)$	12	13	12	7	8	6	20

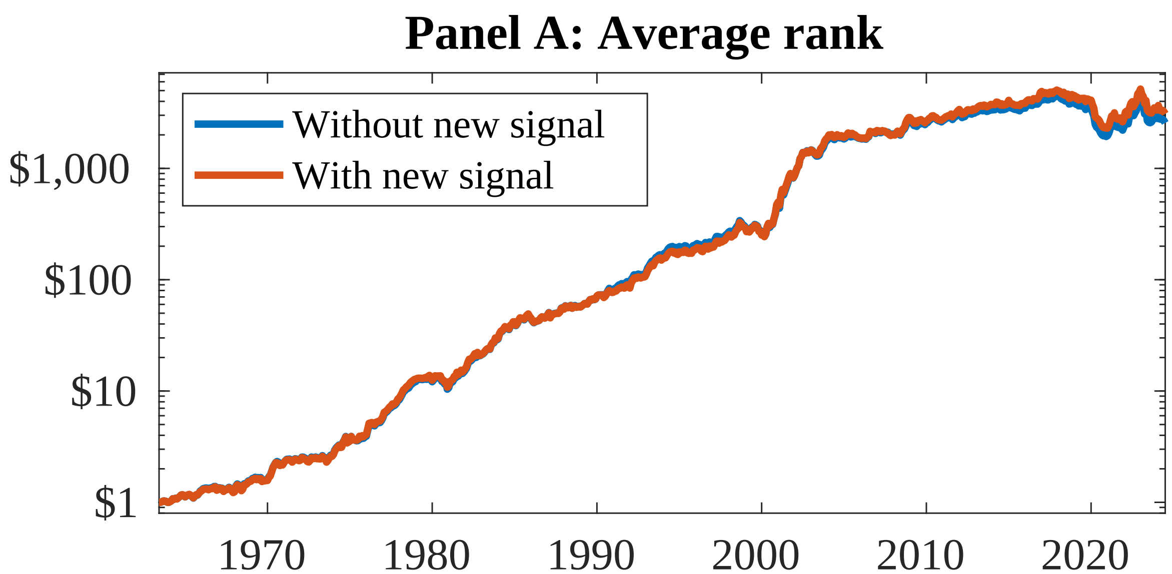


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as PDR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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